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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

MINI PROJECT REPORT

HEALTHCARE DATA ANALYSIS AND DISEASE RISK PREDICTION

Exploratory Data Analysis, Visualization and Data Preprocessing

Submitted by

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Submission Date: 24 October 2026

CERTIFICATE

This is to certify that the mini project report titled “Healthcare Data Analysis and Disease Risk Prediction” is submitted by SHALINI R, with team member SAKTHIS SREE S, from the Department of Artificial Intelligence and Data Science, SRM TRP Engineering College. The project analyzes patient health indicators and uses data preprocessing, exploratory data analysis and visualization to examine patterns across the available disease categories.

The report is submitted as part of the academic mini project work under the guidance of V.Ezhil Swanly.

ACKNOWLEDGEMENT

We express our sincere gratitude to **SRM TRP Engineering College** and the **Department of Artificial Intelligence and Data Science** for providing us with the opportunity, infrastructure, and facilities to successfully complete this mini project.

We express our heartfelt gratitude to our **Dr.Perumalraja, Professor & Head, Department of Artificial Intelligence and Data Science,** for providing constant encouragement, valuable support, and a motivating academic environment that greatly contributed to the successful completion of this project.

We sincerely thank our project guide, **Mr. V. Ezhil Swanly, Assistant Professor, Department of Artificial Intelligence and Data Science**, for his valuable guidance, continuous encouragement, constructive suggestions, and unwavering support throughout the development of this project.

We also extend our sincere thanks to all the faculty members of the Department of Artificial Intelligence and Data Science for their valuable inputs and encouragement. We are grateful to our classmates and family members for their continuous support, cooperation, and motivation during the preparation of the healthcare dataset, data analysis notebook, and project report.

Finally, we acknowledge the healthcare datasets and spreadsheets used for this study, as well as the open-source Python libraries and tools that supported the data preprocessing, analysis, visualization, and report preparation. Their availability greatly contributed to the successful completion of this mini project.

SHALINI R

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ABSTRACT

Healthcare data analysis can help organize patient information and explore patterns among health indicators. This project, “Healthcare Data Analysis and Disease Risk Prediction,” uses a supplied healthcare dataset containing Age, Gender, BMI, Smoking, Glucose, BP, Cholesterol and Disease. Python, Pandas, Matplotlib and Seaborn are used to inspect the data, compare raw and cleaned spreadsheets, and create five visualizations: patient age distribution, average BMI and glucose by disease category, blood pressure versus cholesterol by smoking status and gender, disease distribution across gender, and a correlation heatmap.

The supplied dataset contains 15 patient records and eight columns. The raw spreadsheet contains missing values, while the cleaned spreadsheet has no missing values in the inspected fields. The plots show sample-level variation and associations, including positive correlations between glucose and blood pressure (0.85), BMI and glucose (0.78), and BMI and blood pressure (0.72). This is an educational exploratory project; the charts do not represent a clinically validated diagnosis or a trained predictive model.

CHAPTER 1: INTRODUCTION

1.1 Background

Healthcare datasets include demographic details and measurements such as body mass index, glucose, blood pressure and cholesterol. Systematic data analysis helps make these records easier to inspect and compare. Visualization can reveal distributions and relationships that may deserve further investigation.

1.2 Problem Statement

Raw healthcare records may contain missing values and categorical information that is difficult to analyze directly. The project addresses this by inspecting the raw data, examining the cleaned data, summarizing patient indicators and visualizing their variation across the available disease categories and gender groups.

1.3 Objectives

- Inspect the raw and cleaned healthcare datasets and identify missing values.
- Summarize dataset dimensions, data types and descriptive statistics using Pandas.
- Visualize patient age distribution and compare average BMI and glucose across disease categories.
- Explore blood pressure and cholesterol by smoking status and gender.
- Use a correlation heatmap to examine pairwise relationships and document limitations.

1.4 Scope

The project focuses on data preprocessing and exploratory data analysis of the supplied 15-record sample. It does not establish medical causation, provide clinical advice or demonstrate a trained machine-learning model with validated predictive performance.

1.5 Expected Outcome

The expected outcome is a cleaned spreadsheet, a reproducible Google Colab notebook, visualizations and a summary of sample-level observations that can support further study of healthcare analytics.

CHAPTER 2: DATASET DESCRIPTION

2.1 Dataset Source

The project uses the supplied Excel workbooks: messy_healthcare_data(1).xlsx and cleaned_healthcare_data(1).xlsx. The notebook loads the workbooks with Pandas for inspection and visualization.

2.2 Dataset Overview

- Raw dataset: 15 records and 8 columns
- Cleaned dataset: 15 records and 8 columns
- Raw dataset missing values: Age 2, BMI 2, Cholesterol 1 and Disease 2
- Cleaned dataset missing values: 0 across the eight columns
- Variables: Age, Gender, BMI, Smoking, Glucose, BP, Cholesterol and Disease

2.3 Attribute Description

Attribute	Description	Role
Age	Patient age in years	Numeric feature
Gender	Encoded gender category (0/1)	Categorical feature
BMI	Body mass index	Numeric feature
Smoking	Encoded smoking status (0/1)	Categorical feature
Glucose	Glucose measurement	Numeric feature
BP	Blood pressure value	Numeric feature
Cholesterol	Cholesterol measurement	Numeric feature
Disease	Encoded disease category/label	Target/category label
Disease	Encoded disease category/label	Target/category label

2.4 Dataset Preparation

The raw file uses readable category labels such as Male/Female, Yes/No and disease names. In the cleaned file, Gender, Smoking and Disease are represented by numeric codes. These codes are categorical labels; their precise mapping should be documented in a data dictionary before any predictive modeling.

CHAPTER 3: METHODOLOGY

3.1 Project Workflow

1. Load the raw and cleaned Excel workbooks in Google Colab.
2. Inspect dataset dimensions, column names, data types and missing-value counts.
3. Use the cleaned spreadsheet for descriptive statistics and visualization.
4. Create an age histogram with a kernel density estimate.

5. Compare average BMI and glucose by disease category and plot BP against cholesterol.
6. Plot disease counts by gender and generate a correlation heatmap.
7. Interpret the charts cautiously and record project limitations and future work.

3.2 Tools and Technologies

Python in Google Colab, Pandas for tabular data processing, Matplotlib and Seaborn for visualization, and Microsoft Excel-compatible spreadsheets were used in this project.

CHAPTER 4: DATA PREPROCESSING

4.1 Data Inspection

The notebook reads the two spreadsheets and checks their shape, dataset information, descriptive statistics and missing-value counts. Both the raw and cleaned datasets contain 15 rows and eight columns.

4.2 Cleaning and Preparation

The raw workbook contains missing values in Age, BMI, Cholesterol and Disease. The cleaned workbook has no missing values across its eight columns. The report describes the observed files; any additional imputation or validation method should be documented explicitly in the notebook.

4.3 Categorical Encoding

Gender, Smoking and Disease are encoded as numeric values in the cleaned spreadsheet. Numeric encoding makes plotting and computation easier, but category codes should not be treated as continuous measurements or ordered severity levels unless the dataset documentation supports that interpretation.

4.4 Feature Review

The features reviewed are Age, Gender, BMI, Smoking, Glucose, BP, Cholesterol and Disease. The notebook uses these columns to create group summaries, scatter plots and a correlation matrix.

4.5 Clean Dataset Summary

The cleaned dataset contains 15 records, eight fields and no missing values in the inspected columns. Its small size makes it suitable for demonstrating the analysis workflow, but not for reliable population-level conclusions.

CHAPTER 5: EXPLORATORY DATA ANALYSIS

5.1 Dataset Summary

Metric	Result
Raw dataset records	15
Cleaned dataset records	15
Number of columns	8
Missing values in raw data	Age: 2; BMI: 2; Cholesterol: 1; Disease: 2
Missing values in cleaned data	0 across all columns

5.2 Age Distribution

The age histogram shows observations extending approximately from 35 to 65 years, with the largest bin around the late 40s. The KDE curve provides a smoothed view of the sample distribution; with only 15 records, it should be interpreted descriptively.

5.3 Average BMI and Glucose by Disease Category

The grouped bar chart compares average BMI and glucose for each encoded disease category. In the displayed sample, glucose is highest for category 2. The clinical meaning of each category depends on its documented mapping and cannot be inferred from the numeric code alone.

5.4 Blood Pressure versus Cholesterol

The scatter plot shows blood pressure against cholesterol, with colors for smoking status and marker shapes for gender. Each point represents one patient record. The plot supports visual comparison only and does not demonstrate that one variable causes changes in the other.

5.5 Visualizations

Figure 5.1: Age Distribution of Patients

The histogram and KDE curve summarize the age distribution of the 15 patient records.

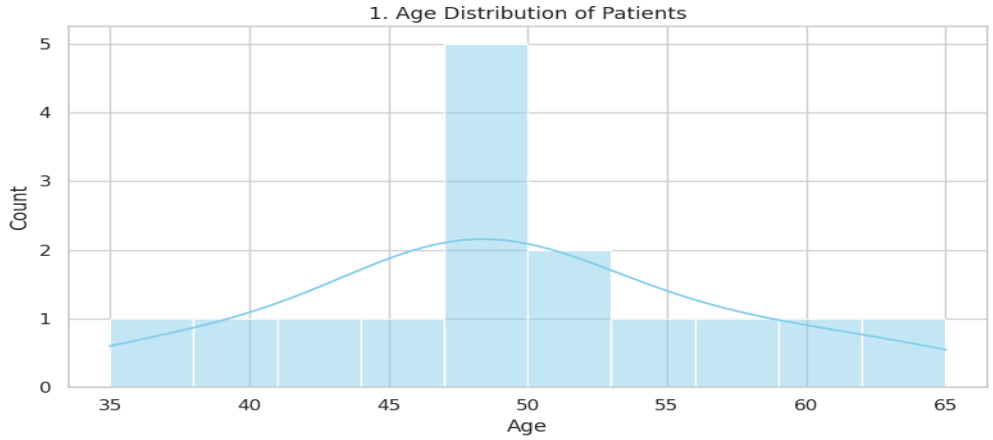


Figure 5.2: Average BMI and Glucose by Disease Category

The grouped bars compare average BMI and glucose values across the encoded disease categories.

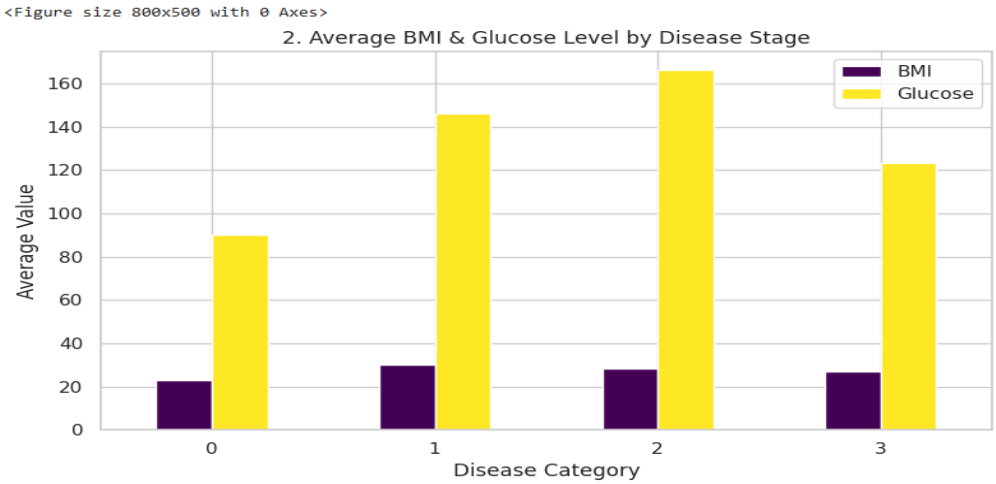


Figure 5.3: Blood Pressure vs Cholesterol

The scatter plot separates records by smoking status and gender for visual comparison.

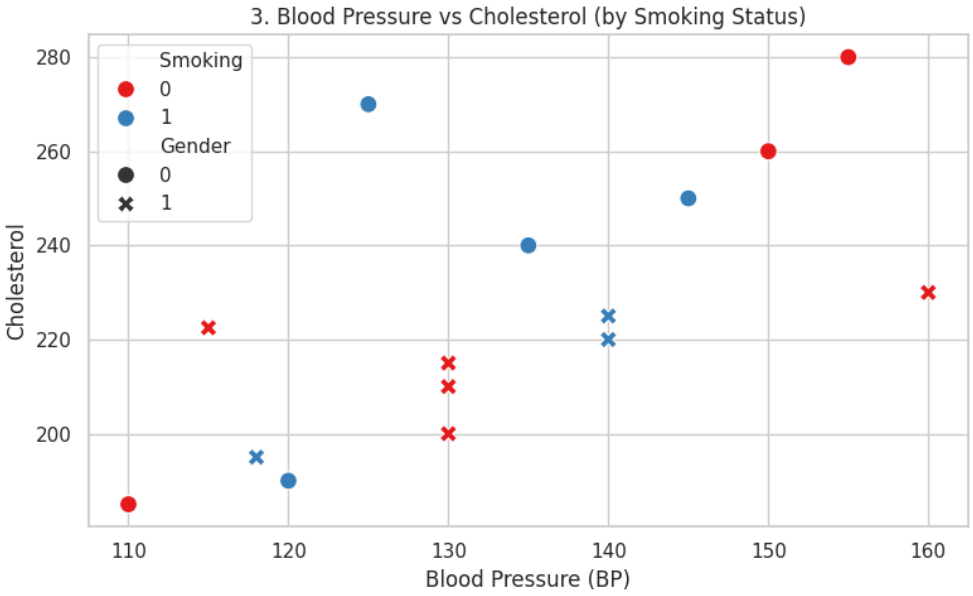
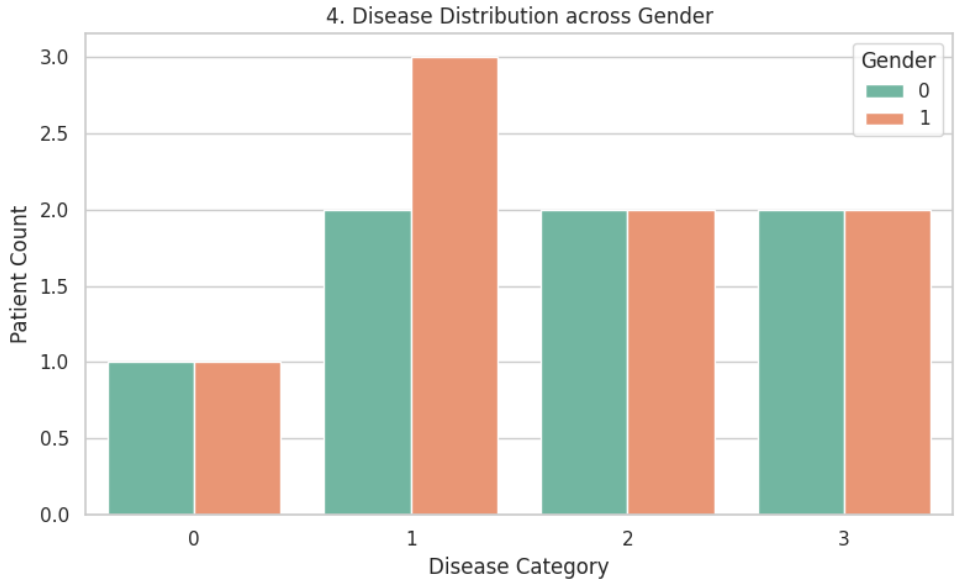


Figure 5.4: Disease Distribution across Gender

The count chart compares the number of records in each disease category for the two encoded gender groups.



CHAPTER 6: RESULTS AND INTERPRETATION

6.1 Disease Distribution by Gender

- The count plot shows that category 1 has more records for gender code 1 than gender code 0 in this sample. The other categories have similar or equal counts in the displayed chart. These are sample counts, not evidence of population-level differences.

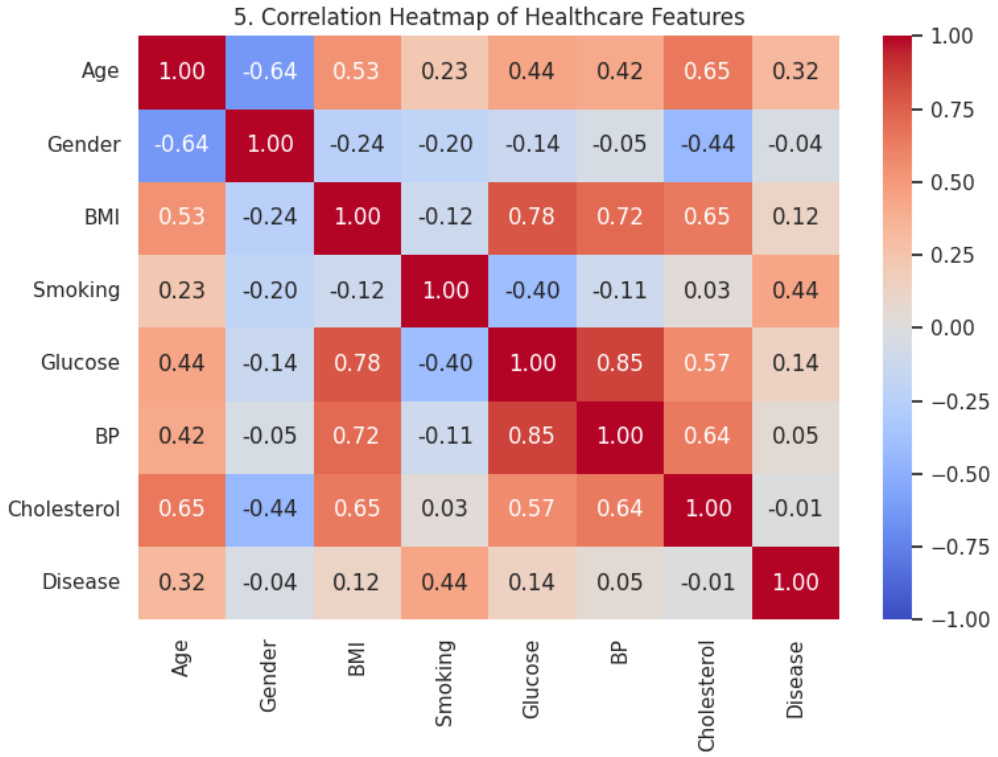
6.2 Correlation Heatmap

- The correlation heatmap displays pairwise correlation coefficients among the numeric columns. The largest positive relationships shown include Glucose with BP (0.85), BMI with Glucose (0.78), and BMI with BP (0.72). Age and Gender show a negative coefficient of -0.64 in this encoded sample; interpretation depends on the category coding.

6.3 Interpretation

- The visualizations make it easier to compare distributions and relationships in the supplied records. Correlation measures linear association, not causation. Because the dataset is very small and category mappings are not fully documented, the results should be treated as exploratory observations only.

6.4 Project Output



The project output consists of the cleaned healthcare workbook, Google Colab notebook and five charts. These outputs demonstrate an end-to-end introductory workflow for healthcare data inspection, preprocessing and visualization.

CHAPTER 7: KEY FINDINGS AND RECOMMENDATIONS

7.1 Key Findings

- The supplied dataset contains 15 patient records and eight variables.
- The raw workbook has missing values; the cleaned workbook has no missing values in the inspected columns.
- The age plot spans approximately 35–65 years, with the highest histogram bin around the late 40s.

- Glucose and BP have a positive correlation of 0.85, while BMI and glucose have a positive correlation of 0.78 in the displayed matrix.

7.2 Recommendations

- Collect a larger, representative dataset with clearly defined variables, units and disease labels.
- Maintain a data dictionary for numeric Gender, Smoking and Disease codes.
- Validate missing-value handling, duplicate records and plausible measurement ranges.
- If prediction is developed later, define the target and train/test separation before modeling.
- Evaluate any future model using suitable metrics such as precision, recall, F1-score and a confusion matrix.
- Obtain appropriate clinical review before considering any real-world healthcare use.

CHAPTER 8: LIMITATIONS AND FUTURE ENHANCEMENTS

8.1 Limitations

- The dataset contains only 15 records, so the results cannot be generalized reliably to a wider patient population.
- Numeric category codes are not fully explained by a data dictionary in the supplied material.
- Correlation values can be unstable in a small sample and may be strongly influenced by individual records.
- The supplied notebook and charts do not demonstrate a trained or evaluated machine-learning prediction model.

8.2 Future Enhancements

- Collect a larger and ethically sourced dataset with reliable labels and documented measurement units.
- Document and validate preprocessing, missing-value treatment, encoding and range checks.
- Train and compare suitable classification models only after defining the prediction target and adequate data are available.
- Use cross-validation and appropriate performance metrics, then review model explanations and risks with qualified professionals.

CHAPTER 9: CONCLUSION

This mini project demonstrates a basic healthcare data analysis workflow using Python, Pandas, Matplotlib and Seaborn. The supplied raw and cleaned workbooks were inspected, and five visualizations were used to explore age, BMI, glucose, blood pressure, cholesterol, gender and disease-category counts. The correlation matrix shows positive associations between glucose and BP and between BMI and glucose in this small sample. These findings are exploratory only. A dependable disease-risk prediction system would require more representative data, documented labels, a properly trained and evaluated model, and qualified clinical oversight.

REFERENCES

- User-supplied healthcare spreadsheets: messy_healthcare_data(1).xlsx and cleaned_healthcare_data(1).xlsx.
- Pandas documentation: https://pandas.pydata.org/docs/
- Matplotlib documentation: https://matplotlib.org/stable/
- Seaborn documentation: https://seaborn.pydata.org/

APPENDIX: PROJECT FILES AND VISUALIZATIONS

The project materials used for this report include the Google Colab notebook, raw and cleaned healthcare spreadsheets, and the visualization outputs.

- heathcare data analysis.ipynb
- messy_healthcare_data(1).xlsx
- cleaned_healthcare_data(1).xlsx
- Healthcare_Data_Analysis_Disease_Risk_Prediction_Report(1).pdf
- kernal density estimate(1).png — Age distribution
- grouped bar chart(1).png — Average BMI and glucose by disease category
- scattter plot(1).png — Blood pressure vs cholesterol
- grouped bar chart (1).png — Disease distribution across gender
- correlation heatmap(1).png — Correlation matrix

GITHUB LINK:

https://github.com/ammushalini-hue/Healthcare-Data-Analysis-and-Disease-Risk-Prediction